

Advanced Programming – Exam 10 Jul 2026 - Part 1

Name:

Question 1 (1.2 points)

What is the difference between a class and an object? Give one example.

Question 2 (1.2 points)

What is the difference between `std::vector` and a C-style array?

Name:

Question 3 (1.2 points)

What is polymorphism? What is the difference between compile-time and run-time polymorphism?

Question 4 (1.2 points)

What is the purpose of a virtual destructor? Give an example where omitting it leads to incorrect behavior.

Name:

Question 5 (1.2 points)

Explain the difference between passing an object by value, by reference, and by const reference. When would you use each?

Question 6 (1.2 points)

Explain the difference between mutable and immutable objects in Python. Give two examples of each.

Name:

Question 7 (1.2 points)

Write a Python function that receives a list of numbers and returns a new list containing only the even values using a list comprehension.

Question 8 (1.2 points)

Using NumPy, explain the difference between element-wise multiplication and matrix multiplication.

Name:

Question 9 (1.2 points)

Using NumPy, generate a vector of 100 equally spaced values in the interval $[0, 1]$ and compute its Euclidean norm.

Question 10 (1.2 points)

Suppose you have implemented a C++ library. Briefly describe the steps required to make it available from Python using pybind11.